جامعة أم القرى
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College of Computing

Computer Science and Artificial Intelligence

AI-Driven Marketing Intelligence Using Persona-Aware Engagement Prediction

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ABSTRACT

Digital marketing systems often address audience analysis, content generation, and engagement prediction as separate tasks, limiting their ability to evaluate how structured audience modeling influences predictive performance. This project proposes an AI-driven marketing intelligence framework that experimentally integrates persona mining, content generation, and engagement prediction into a unified pipeline. Audience personas are constructed using sentence embeddings generated by SBERT and clustering techniques such as K-Means and Hierarchical Clustering. These persona-aware representations are then incorporated into an engagement prediction model to examine whether they provide measurable improvements over text-only baselines. A baseline model is developed using textual and statistical features, while an enhanced model integrates persona-based features including cluster identifiers and persona-level engagement statistics. Controlled experiments and ablation studies are conducted to compare performance using quantitative metrics such as RMSE, Accuracy, and F1-score. The expected outcome is an experimental research prototype that demonstrates whether persona-aware modeling significantly improves engagement prediction accuracy within an integrated AI-driven marketing workflow.

Keywords: Persona Modeling, Engagement Prediction, Clustering, SBERT

1. BACKGROUND AND CONTEXT

In digital marketing, understanding customer behaviour is an important step before creating marketing content. Many companies analyse customer reviews, comments, and social media posts to identify user interests and preferences. Natural Language Processing (NLP) techniques such as sentiment analysis are commonly used to extract opinions and emotions from text data, which helps in building customer personas and improving marketing decisions (Bolla, 2024). Some studies also use clustering and recommendation techniques to group customers based on similar behaviours or preferences. By analysing textual reviews, researchers can discover patterns in customer needs and create more personalised marketing strategies (Danish & Amjad, 2023). Other research focuses on extracting user interests from social media platforms like Twitter to understand audience lifestyles and habits, which can support persona modelling for targeted advertising (Mohamed et al., 2023). These studies show that persona mining using NLP and machine learning can help businesses understand their audience better. However, most research focuses only on analysing customer data without connecting the results to content generation or engagement prediction in one integrated system.

In digital marketing, creating attractive and persuasive content is an important step to reach the target audience. Many studies use Artificial Intelligence and Natural Language Processing (NLP) techniques to automatically generate marketing text such as advertisements, captions, and product descriptions. Large language models developed by organizations such as OpenAI and Google are widely used to produce fluent and context-aware marketing content (Loukili et al., 2024). Some research focuses on personalized content generation by using user features such as demographics or behaviour data to improve marketing strategies (Pasupuleti, 2024). Other studies explore how AI models can support digital advertising through automated content creation and campaign optimization (Pandey, 2025). In addition, research on generative AI shows how advanced NLP models can improve text generation quality and adaptability in marketing applications (Salian, 2025). These studies show that AI-based content generation can help marketers create fast and effective content. However, most research focuses only on generating content without adapting it to different social media platforms or connecting it to persona-based analysis and engagement prediction in one integrated system.

In digital marketing, predicting user engagement is essential for evaluating campaign effectiveness and is commonly measured using metrics such as likes, shares, comments, and click-through rates (CTR). Many studies apply machine learning techniques, including regression, classification, and deep learning models such as neural networks and transformers, to predict engagement based on textual, behavioural, contextual,

user, and temporal features (Kumar & Singh, 2023; Li et al., 2024). By extracting semantic representations from marketing text and combining multiple feature types, these models better capture the relationship between content characteristics and user interaction, leading to significantly improved prediction accuracy compared to text-only approaches (Rahman & Chen, 2023). Although engagement prediction using AI supports data-driven marketing decisions, most studies treat it as an isolated task and do not examine the impact of persona-based audience modelling within an integrated marketing framework.

Several studies have explored the use of Artificial Intelligence in digital marketing. For example, Pasupuleti, M. (2024) discussed how AI can improve personalized marketing strategies by using large language models to generate marketing content based on customer preferences. The study showed that AI helps companies create more targeted and effective promotional messages. Another study by Yang, Q. et al. (2025) proposed combining predictive models with large language models to improve marketing recommendations. Their system used customer data to generate marketing ideas and then predict performance before publishing content, which helps companies make better marketing decisions. These studies show that AI can be used to generate content and support marketing decisions. However, most research still focuses on specific parts of the process, such as content generation or prediction alone. There is still limited work on building a fully integrated system that connects audience persona mining, platform adaptive content generation, and engagement prediction in one continuous workflow, which is what our project aims to achieve.

2. PROJECT SCOPE

The scope of this project focuses on designing and developing an AI-driven marketing intelligence platform that supports marketers in understanding their target audience, generating customized marketing content, and predicting content performance before publishing. The project will address the analysis of textual data such as customer reviews, social media comments, and marketing-related posts. Based on this data, the system will extract audience characteristics, generate marketing content tailored to different audience personas and adapted to different social media platforms, and estimate expected engagement metrics. The scope of the project is limited to text-based digital marketing content and does not include image, video, or real-time trend analysis. In addition, the system relies on historical data and predefined datasets for training and evaluation. External factors such as sudden viral trends or real-world events are outside the scope of this project.

3. PROJECT DESCRIPTION AND IDENTIFIED GAP

Current AI-based marketing solutions commonly address isolated tasks such as sentiment analysis, automated text generation, or engagement prediction. Although these components are technically advanced, there is limited empirical investigation into how structured audience modeling influences downstream predictive performance. In particular, most existing systems do not quantitatively evaluate whether incorporating persona-aware representations improves engagement prediction accuracy compared to models relying solely on raw textual features. Based on the reviewed literature, existing AI-driven marketing systems typically address audience analysis, content generation, and engagement prediction as separate research tracks. While some studies explore persona construction using clustering techniques, others focus on automated content generation or predictive modeling independently. However, there is limited evidence of a unified framework that experimentally integrates these three components within a single closed-loop system and evaluates their combined impact on predictive performance. The technical gap addressed in this project is the absence of measurable analysis regarding the contribution of persona-based features to marketing performance prediction. Specifically, this project investigates whether integrating audience personas constructed using sentence embeddings (e.g., SBERT) and clustering techniques enhances engagement prediction models. The improvement will be experimentally validated using quantitative metrics such as RMSE for regression and Accuracy and F1-score for classification. Furthermore, the project compares different persona construction methods (e.g., K-means versus Hierarchical Clustering) and evaluates their impact on both generated content quality and prediction performance. Through controlled experiments and ablation studies (with and without persona features), the system aims to determine whether persona-aware modeling provides statistically significant improvements

within an integrated AI-driven marketing framework.

4. PROJECT OBJECTIVES

• Implement audience persona modeling using sentence embeddings (e.g., SBERT) combined with clustering algorithms such as K-means and Hierarchical Clustering.
• Evaluate the quality of the constructed personas using clustering validation metrics (e.g., Silhouette Score) and interpretability analysis.

• Develop a baseline engagement prediction model using textual features only.

• Extend the prediction model by incorporating persona-aware features and compare performance using quantitative metrics such as RMSE, Accuracy, and F1-score.

• Conduct an ablation study to measure the impact of persona-based features on prediction performance and determine statistical significance of improvements.

5. PROJECT TIMELINE

Week	Activity
1	Topic selection and initial discussion
2	Initial literature review
3	Problem definition and scope refinement
4	Initial Methodology
5	Proposal writing, revision, and submission
6-8	Methodology planning and pipeline
9-11	Dataset selection
12-14	Preliminary evaluation
15-16	GP1 report preparation

6. EXPECTED OUTCOMES

This project is expected to deliver an experimental research prototype that studies the effect of persona-aware modeling on marketing content generation and engagement prediction. The outcomes will include a preliminary system framework that connects persona construction, content generation, and engagement prediction in one experimental pipeline, along with an initial implementation for building audience personas and generating tailored marketing content. The engagement prediction model will be evaluated through controlled experiments using RMSE and MAE for regression tasks and Accuracy and F1-score for classification tasks (such as high vs. low engagement). The results will also compare models with and without persona features, as well as different persona construction methods, to check whether persona-aware modeling provides measurable improvements in prediction performance.

7. PROPOSED METHODOLOGY OR APPROACH

In this project, a controlled experimental methodology is adopted to examine the technical impact of persona-aware features on engagement prediction. The process begins with data collection, where a dataset of marketing posts is gathered from publicly available sources, and each post includes textual content and a numerical engagement value, followed by splitting the dataset into training and testing sets for fair evaluation. Text Preprocessing and Representation: text preprocessing includes cleaning, lowercasing, removing special characters, tokenization, and stop-word elimination, while Sentence-BERT (SBERT) is used to generate dense sentence embeddings that capture semantic meaning. Persona Mining: audience personas are constructed by applying K-Means clustering to the SBERT embeddings due to its efficiency

and scalability, with the optimal number of clusters determined using the Elbow Method and validated through the Silhouette Score to ensure quantitative support. Baseline Engagement Prediction Model: a baseline model is implemented using a Random Forest Regressor based on textual and statistical features such as TF-IDF, text length, sentiment polarity, and numerical metadata, and evaluated using RMSE and MAE. Persona-Aware Engagement Prediction Model: an enhanced model is then developed using the same regressor with the addition of persona-aware features, including cluster ID, distance to the cluster centroid, and average engagement statistics per persona, allowing performance improvements to be attributed to persona-based information rather than changes in model structure.

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